Yinghao (Peter) Guan

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Yinghao-Guan | yinghao-guan

Education

Santa Monica College, Santa Monica, California — Sept 2025 – Present
Computer Science
- Dean's Honor List; Alpha Gamma Sigma; CS Club; HackSMC

University of Edinburgh, Edinburgh, Scotland — Sept 2022 – July 2024
Computer Science (Completed 2 Years)
- Related coursework: Software Engineering, Reasoning and Agents, Data Structures, Computer Systems, Logic

University of Auckland, Auckland, New Zealand — Feb 2022 - Jun 2022
Finance & Engineering (Completed Coursework)
- Relevant coursework: Business, Economics, Markets, and Law

Technical Skills

- **Frontend:** React, Next.js, Tailwind CSS, TypeScript, JavaFX, Tauri
- **Backend & Cloud:** FastAPI, Flask, PostgreSQL, Firebase, Vercel, Render
- **Languages & Data:** Rust, C++, Java, Python, TypeScript, R, SQL, pandas, NumPy, scikit-learn, Stata
- **Tools:** Git/GitHub, Docker, GitHub Actions, Linux, DNS/domain management

Experience

HCI Research Collaboration | *Lead Student Researcher* — Jul 2025 – Present
UCLA, Los Angeles, California
- Investigating the intersection of **Explainable AI (XAI)** and user trust, specifically focusing on emotional support LLMs.
- Designed and executed A/B testing frameworks to evaluate user empathy and trust across different AI transparency levels.
- Automated qualitative data extraction and sentiment analysis using Python, preparing for peer-reviewed publication.

Website Coordinator | *Alpha Gamma Sigma Honor Society* — Feb 17, 2026 – Present
Santa Monica College, Santa Monica, California
- Maintain the official society website and manage feature updates to keep the member experience reliable and current.
- Publish announcements, event schedules, and resources through the organization's main digital communication channel.
- Administer the digital member management system for onboarding, engagement tracking, and roster accuracy.

Veru: AI-Driven Academic Audit Platform (veru.app) | *Founder* — Dec 2025 – Present
- Engineered a full-stack platform to automate academic paper auditing, leveraging a **FastAPI** backend and a responsive **Next.js** frontend.
- Developed a data pipeline integrating **Semantic Scholar** and **OpenAlex** APIs, using LLM-based verification to cross-reference citations and findings in real time.
- Successfully launched the MVP, achieving **200–300 monthly unique users** with **daily active usage** and consistent API traffic.
- Optimized API performance through asynchronous request handling, maintaining sub-200ms response times.

Technical Projects

Acadown – Academic DSL & Cross-Platform IDE Ecosystem — Aug 2025 – Present
GitHub
- Designed **Acadown**, a specialized **Domain-Specific Language (DSL)** for academic document authoring, featuring a custom Markdown-like syntax parsed via **ANTLR4** with full grammar specification.
- Architected an **AST-to-Pandoc** transpilation pipeline using the Visitor pattern, enabling multi-format output (HTML, LaTeX/PDF) from a single source document.
- Built a cross-platform desktop IDE for Acadown using **Rust** and **Tauri** with an **MVVM** architecture, featuring live preview and syntax highlighting.

Ultra Marathon Runner Performance Analysis | *Independent Data Analyst* — Jan 2024 – Apr 2024
Foundation Data Science, University of Edinburgh, Edinburgh, UK

- Analyzed a Kaggle dataset of over 200,000 ultra-marathon records to identify long-term performance trends and athlete segments.
- Executed the workflow in Python using **pandas**, **matplotlib**, and **scikit-learn**, including **K-means** clustering across race categories.
- Compiled findings and visualizations into a professional report using LaTeX.

Leadership & Competitions

BroncoHacks 2026 | *Best Sport & Fitness Track Winner* Apr 2026
BrancoHacks, Cal Poly Pomona, CA
GitHub | Live Demo | Devpost

- Trained a multi-output **RandomForestRegressor** (200 trees) on real athlete data augmented with synthetic movement features, producing readiness, strength, endurance, and injury-risk percentiles from 23 inputs.
- Engineered the **FastAPI** backend with SQLite persistence, exposing ML inference, SHA-256 workout proof hashing, and badge eligibility logic through a REST API.
- Authored a **Rust + Anchor** Solana program with four PDA-backed instructions for on-chain athlete profiles, immutable training proofs, and live NFT badge minting via SPL Token on devnet.

Hackonomics Hackathon | *Honorable Mention* Mar 2026
Online
GitHub | Live Demo | Devpost

- Developed **Village-Economic**, a visual novel-style game designed to teach core economic theories through interactive storytelling.
- Built the full application using **Next.js**, implementing game logic, UI flow, and state management to deliver a seamless browser-based experience.
- Collaborated in a **2-person team**, integrating teammate-provided economic content, narrative design, and visual assets into a cohesive product.
- Translated abstract economic concepts into engaging gameplay mechanics, enhancing accessibility and educational impact.

Build with Gemini Hackathon (UCLA x Google DeepMind) | *Runner-Up Prize Winner* Mar 2026
UCLA Glitch, Los Angeles, California
GitHub | Live Demo | Devpost

- Built **Marketeer**, an end-to-end AI marketing campaign generator that produces logos, banners, jingles, video ads, and voiceovers from a single prompt.
- Designed and implemented an asynchronous multi-stage pipeline coordinating **5+ Gemini models** to enable parallel asset generation.
- Engineered a browser-first workflow with IndexedDB persistence and client-side media merging using **FFmpeg WASM**.

Vision Hack | *Top 3 Overall* Mar 2026
Vision Hack, Los Angeles, California
GitHub | Live Demo | Devpost

- Developed the backend for **Skillset LA**, an AI-powered career roadmap builder.
- Integrated Gemini and Adzuna APIs into the application workflow.

Hack for Humanity 2026 | *Solaura* Mar 2026
Santa Clara University, California
GitHub | Devpost

- Led a 4-person team to build a real-time spatial perception system designed to augment independence for visually impaired users.
- Developed the **iOS** application using **Swift** and ARKit, integrating **CoreML YOLOv8** for low-latency object detection.
- Engineered a lightweight **Python** backend for audio synthesis and state synchronization, and integrated **Gemini** and **ElevenLabs** for multimodal voice feedback.

HackCC Fall 2025 Hackathon | *Best AI/ML Prize Winner* Nov 2025
HackCC, MiraCosta College, Ocreanside, California
GitHub | Live Demo | Devpost

- Spearheaded the development of **RealiBuddy**, an AI-driven learning tool with real-time sentiment analysis and adaptive feedback.
- Orchestrated the technical workflow for a 4-person team and ensured seamless integration between the ML model and web interface.
- Implemented a visualization dashboard to provide users with actionable insights into learning progress.